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MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, MULTISTAGE, VERTICAL, ELECTRIC-MOTOR-DRIVEN (FOR LIQUID HYDROCARBON FUELS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a vertical, multistage, centrifugal pump driven by a vertical, hollow-shaft, explosion-proof electric motor.

1.2 Classification. Pumps shall be of the following capacities, as specified (see 6.2):

150 gallons per minute (gpm) at 390 foot total discharge head.
300 gpm at 390 foot total discharge head.
600 gpm at 390 foot total discharge head.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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SPECIFICATIONS

FEDERAL

WW-F-406	- Flanges, Cast-Iron (Classes 125 and 250) and Bronze (Classes 150 and 300).

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-710	- Pump, Deep-Well, Packaging of.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-A-7021	- Asbestos Sheet, Compressed, for Fuel, Lubricant, Coolant, Water, and High Temperature Resistant Gaskets.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.

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MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS49006	- Plugs, Pipe, Magnetic-Headless, Aluminum, Iron or Steel.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.5 - Steel Pipe Flanges, Flanged Valves, and Fittings.
B40.1 - Gages - Pressure and Vacuum, Indicating Dial Type Elastic Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A268 - Seamless and Welded Ferritic Stainless Steel Tubing for General Service.
B584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017).

AMERICAN WELDING SOCIETY (AWS)

B3.0 - Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125).

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

HS1086 - Unified Numbering System for Metals and Alloys.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pumps shall be furnished complete with pump base and electrical motor. The pumps shall be suitable for pumping liquid hydrocarbon fuels. The pumps shall be designed and constructed to withstand hard usage and storage in the open air for extended periods of time.

3.2 First article. The contractor shall furnish one or more pumps as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce pumps that comply with the requirements of this specification. Examination and tests shall be as specified in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examinations and tests, as specified (see 6.2).

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3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.7).

3.3.1 Nonsparking metals. Nonsparking metals shall be used for those parts of the pump where service involving the use of tools may be required when the pumping system is filled or partially filled with petroleum products.

3.3.2 Unified numbering systems (UNS). Where applicable, metal alloys are identified by UNS in accordance with SAE HS 1086.

3.4 Threads. All screw threads shall conform to FED-STD-H28.

3.5 Environmental conditions.

3.5.1 Operating temperature. The pumps shall perform as specified in any ambient temperature from plus 125° F to minus 25° F.

3.5.2 Storage temperature. The pumps shall not be damaged by storage in any ambient temperature from plus 160° F to minus 50° F.

3.6 Safety. All rotating, reciprocating, exposed and electrically energized parts that are of such nature or located so as to become a hazard to operating personnel shall be fully guarded or enclosed. Protection devices shall not impair the operating functions of the pump. Nonfunctional sharp edges, projecting points and excessive lengths of fasteners shall be avoided.

3.7 Human factors. The pump shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472.

3.7.1 Noise limits. The noise of the pump shall not exceed the limits specified in MIL-STD-1474, category D, for steady-state noise when tested in accordance with 4.5.2.6.1. The sound level shall comply with the exterior noise requirements of MIL-STD-1474 when tested in accordance with 4.5.2.6.2 (aural nondetectability and impulse noise do not apply).

3.8 Reliability. The specified mean-time-between failure (MTBF) of the pump shall be 140 hours when tested as specified in 4.5.2.7.

3.9 Maintainability.

3.9.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.4 and 6.5) shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.8.1.

3.9.2 Maintenance ratio. The pump shall have a maintenance ratio of not more than 0.050. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total

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operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.9.3 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments.

3.10 Pump. The pump shall be the multistage, vertical, centrifugal type for installation on horizontal underground storage tanks. The pipe column and strainer shall be removable through a flanged tank sleeve. The distance from the bottom of the pump mounting flange to the bottom of the strainer shall be as specified (see 6.2).

3.10.1 Hydrostatic pressure. The pump shall withstand an internal hydrostatic pressure of not less than 350 pounds per square inch (psi) without leakage, failure, or permanent deformation.

3.10.2 Surface-discharge head. The surface discharge head shall be either ductile iron conforming to UNS F32800; cast iron conforming to UNS F12101; cast steel; or steel. The discharge head shall consist of a finished mounting flange; finished discharge flange; and necessary bolts, gaskets, and fittings for mounting the discharge casting on the tank flange, for attaching the column assembly to the discharge head, and for mounting the motor on the discharge head. The discharge head shall provide sufficient clearance for the adjustment or removal of the pump shaft seal without removal of the motor from the base.

3.10.2.1 Mounting flange. Each pump shall be provided with a mounting flange either ductile iron conforming to UNS F32800 or cast iron conforming to UNS F12101, unless a particular material is specified (see 6.2). The mounting flange shall be either a flange fabricated integrally with the discharge head or a separate flange fitted to the bottom of the discharge head, as specified (see 6.2). The lower face of the flange or adapter ring shall be plain faced and shall be suitable for mounting on the flange of an underground tank. A full-faced flange gasket shall be provided between the pump base or adapter ring and the flange of the underground tank.

3.10.2.2 Discharge flange. Each pump shall be provided with a discharge flange faced and drilled in conformance to WW-F-406 for cast iron or ANSI B16.5 for steel. The flange shall be the following nominal sizes:

Pump capacity	Nominal flange size
150 gpm	4-inch by 9-inch
300 gpm	4-inch by 9-inch
600 gpm	6-inch by 11-inch

3.10.2.3 Flange gaskets. Gaskets shall be 0.062 inch nominal thickness and shall be material conforming to MIL-A-7021 class I.

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3.10.2.4 Surface-discharge, open bypass. A nonadjustable, open bypass which will allow discharge of a minimum of 2 percent of the rated capacity of the pump shall be built into the discharge head to provide cooling liquid circulation at shutoff head conditions. A means of returning the circulating liquid back to a point immediately below the first stage impeller shall be provided.

3.10.3 Shaft. All line and head shafting shall be turned, ground, and polished steel conforming to UNS G10180, cold-finished. At all points where the line and top sections of the shaft pass through the seal or a bearing, the shaft shall be provided with sleeves that are securely fastened. The sleeves shall be of chrome plated or corrosion-resistant steel conforming to ASTM A268, grade TP410. The shaft shall be true and straight in a manner to insure smooth operation without vibration of whip at the operating speed specified herein. The ends of the shaft shall be threaded and faced to insure true and straight operation when connected face to face in the shaft coupling.

3.10.3.1 Shaft couplings. The shaft coupling shall be steel conforming to UNS G10180, cold-finished; and the couplings shall be threaded to fit the shaft. The torsional strength of the couplings shall be not less than the torsional strength of the shaft. Both ends of the couplings shall be counterbored with the inside of the counterbore not less than the full outside diameter of the threads.

3.10.4 Pump-shaft seal. The pump shall be fitted with a mechanical or slinger-type seal.

3.10.4.1 Mechanical-type seal. The mechanical-type seal shall effectively contain an internal hydrostatic pressure of not less than 350 psi without leakage, failure, or permanent deformation of any of its components. The seal shall be self alining and pressure balanced and shall be removable without removal of the motor from the pump base. The adjustment of the sealing surface shall be independent of the impeller adjustment. A carbon-ring insert providing a sealing surface of carbon against super-smooth stellite or equal shall be used. Carbon-ring insert material shall be resin impregnated. In lieu of the carbon surface against supersmooth stellite, properly lubricated, dissimilar, noncorrosive metals of different hardness properties may be utilized. All parts of the mechanical-type seal shall be resistant to corrosion and oxidation. The seal shall provide a smooth, exposed surface which will facilitate relapping. Gaskets made of synethetic Buna-N rubber compound or other materials impervious to the deleterious action of aromatic petroleum products shall be used where a sealing means is required between the pump shaft and the rotating elements of the seal. The rotating elements of the seal shall be fastened to the pump shaft by means of setscrew, pin, or lug. Spacing gage shall be furnished for properly positioning the seal on the shaft. A non-adjustable means shall be provided for cooling the seal with the liquid delivered by the pump.

3.10.4.2 Slinger-type seal. The slinger-type seal shall be designed in such a manner that external leakage at the pump-shaft shall be eliminated by pressure breakdown in the column case. The seal shall be provided with a long shaft

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support bearing mounted in the discharge column, a slinger attached to the shaft above the bearing, vapor seal cup, seal gasket, seal insert, O-rings, a bypass port, and a shaft sealing tube which extends from a second bearing, mounted above the slinger, to the discharge head. The shaft-support bearing shall be long enough to limit the volume of leakage so as to make the slinger effective. Any liquid which may leak past the shaft-support bearing shall be diverted by the slinger through the bypass ports back into the storage tank. The shaft sealing tube shall enclose the shaft between the upper bearing and the discharge head and shall prevent leakage at the point where the shaft passes through the discharge head. The O-ring-type-vapor-seal shall prevent leakage of vapor at the point where the shaft passes through the discharge head.

3.10.5 Pipe column. The pipe column shall be steel conforming to UNS K03005, and shall extend from the pump base to the top bowl of the pump. All connections including the pump top bowl, may be either flanged or threaded. The flange, if used, shall be double welded to the pipe. If straight-thread connections are used, the threads on the pipe and the end of the pipe sections shall be faced so as to butt on the end of the spider ring in the coupling and shall form a tight joint. The pipe-column sections shall be not more than 10 feet in length, and all pertinent parts shall be machined to insure accurate centering and alinement of the sections, shafting, bearings, and pump bowls.

3.10.6 Line-shaft bearings. The line-shaft bearings shall be mounted in a spider which will offer a minimum resistance to the moving column of pumped liquid. For a threaded column, the outer rings of the spiders shall be threaded and secured in the column pipe between the butt ends of the column pipe. The bearings shall function satisfactorily when lubricated and cooled with the pumped liquid. The bearings shall be bronze conforming to UNS C93700, C93800, or C94300.

3.10.7 Pump bowls. The pump bowls shall be cast iron conforming to UNS F12101, or ductile iron conforming to UNS F32800. The bowls shall have smooth interior surfaces and liquid passages. Each bowl shall be fitted with not less than one bearing made of aluminum bronze conforming to UNS C95400. When a discharge case is used as a separate piece, an additional bearing shall be mounted in the discharge case immediately above the top bowl, and another bearing shall be mounted in the suction manifold. The bowls may be either threaded or flanged.

3.10.8 Impellers. Unless otherwise specified (see 6.2), the impellers shall be bronze conforming to UNS C92700. The impellers shall have smooth liquid passages and machined outside surfaces. The bottom impeller shall be securely locked to the impeller shaft by means of a split taper lock bushing. Impellers shall be either semi-open or enclosed type. Enclosed type impellers shall be provided with wearing rings or seals for running closure with the pump bowls. Wearing rings shall be high-lead tin bronze conforming to ASTM B584. Rings shall be a tight press fit on the impeller eye or in the pump bowl. Seal rings shall be Buna-N rubber stiffened with a copper or bronze ring.

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3.10.9 Impeller shaft. Unless otherwise specified (see 6.2), the impeller shaft shall be corrosion-resistant steel conforming to UNS S31600.

3.10.10 Suction fitting. The suction of the pump shall be flared, and the bearing spider shall be streamlined to reduce entrance losses. A corrosion-resistant steel grating or heavy wire mesh of a size to prevent the entrance of solids that would damage or clog the pump shall be fitted to the suction end. The lower pump-shaft bearing shall be zinc-free, high-lead bronze.

3.10.11 Pressure gage. A pressure gage conforming to ANSI B40.1, grade A, shall be installed at the discharge flange of each pump. The gage shall have a brass, enclosed, 4-1/2-inch diameter dial and shall be provided with a bronze bar stock needle valve. The range of the gage shall be from zero to not less than 300 psi.

3.11 Electric motors. The electric motor shall conform to NEMA MG-1, poly-phase type, squirrel-cage induction, hollow-shaft, totally enclosed, fan-cooled, vertically mounted, explosion-proof type. The rating of the motor shall be based upon 55° C temperature rise. The motor shall be suitable for class I, group D, hazardous installations in accordance with NFPA 70. Insulating material of the motor shall be class B. The motor shall operate with a counterclockwise rotation when viewed from the top. An arrow shall be cast or stencilled on the top of the drip cover indicating the direction of rotation.

3.11.1 Nonreverse, rachet-type mechanism. The motor shall have a built-in, nonreverse, rachet-type mechanism that shall prevent reversal of the pump shaft if the motor is started in the reverse direction.

3.11.2 Power rating. The continuous duty rating of the motor shall be not less than the required pump horsepower at rated power requirement conditions (see applicable rating in 1.2).

3.11.3 Speed. The synchronous speed of the motor shall be 1,800 revolutions per minute (rpm).

3.11.4 Voltage and frequency. Unless otherwise specified (see 6.2), the motor shall operate on a 440-volt, 3-phase, 3-wire, 60-Hertz, electrical power system.

3.11.5 Motor shaft. The motor shall be equipped with a top pump driver shaft which permits adjustment of the impeller clearance from the top of the motor by means of a top shaft adjusting nut. The shaft shall be equipped with a coupling below the lower motor bearing, located and positioned to provide sufficient clearance for adjustment or removal of the mechanical-type pump seal (when the mechanical-type seal is provided) without the removal of the motor from the base. The coupling shall withstand the vertical thrust. The coupling shall be either threaded or spline-sleeved with smooth exterior surfaces and shall be provided with a positive means for removal by use of a spanner wrench. The top pump driver shaft and coupling shall be corrosion-resisting steel conforming to UNS S31600.

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the flexible coupling.

3.11.6 Motor bearings. Anti-friction motor bearings including a topthrust bearing shall be provided. The bearings shall function on a continuous duty basis while supporting the maximum combined radial and thrust loads imposed by the weight of the pump rotating elements and the hydraulic action of the pump.

3.12 Performnce. The pump shall deliver not less than the capacity in gallons per minute specified in 1.2 of liquid hydrocarbon fuels having specific gravities of from 0.68 to 0.935 at temperatures from 40° F to 80° F. The ratio between the shut off head and total head at peak efficiency shall be within the limits of 1.25 to 1.35. The pump shall have an efficiency of not less than 70 percent at rated discharge head and capacity.

3.13 Electromagnetic interference. When specified (see 6.2), the electromagnetic interference emission and susceptability characteristics of the pump shall conform to MIL-STD-461, class IIB.

3.14 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.14.1 Lubricants. The pump shall operate as specified herein when lubricated with military lubricants.

3.14.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.14.4 Fillings, draining, and checking provisions. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gauges to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drainplug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accomplished when the pump is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet. The drainplug, the filling means, and the lubricant level dipsticks or holes shall be accessible without disturbance of accessories or parts.

3.14.5 Grease lubrication. All grease lubrication, including the lubrication of sealed bearings, shall be with greases as follows:

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a. MIL-G-23827 - For use on gears, sealed bearings, actuator screws, and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.
b. MIL-G-10924 - For lubrication and surface corrosion protection of bearings operated over the temperature range of minus 65° F to plus 225° F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable.

The pump shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.14.6 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.15 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.15.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.16 Lifting provisions. Permanently affixed attachments that enable the pump to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the pump, 2.5 times the resulting vectorial force imposed when the pump is suspended by the lifting attachments. Attachments for multiple slings shall be located so that: (a) slings will converge at not more than 24 feet above the lowest extremity of the

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pump; and (b) not less than 1-inch clearance will be maintained between the slings and the pump. Eye openings and clearance dimensions of the attachment eye shall conform to MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.17.

3.17 Tiedown provisions. When specified, permanently affixed attachments that permit fastening the pump to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the pump, its proportion of the following static loads: 8-g (acceleration due to gravity) forward, 8-g rearward, 2.25-g upward, 1.5-g sideward, where g equals cargo weight of the pump. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.16.

3.18 Identification marking. The pump shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump on identification plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.19 Treatment and painting. The portions of the pump normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.20 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

3.21 Workmanship. The pump shall be free from defects such as misalined components, incomplete welds, cracks, and any other defects that could impair the operating and serviceability characteristics of the pump. All piping, fitting, hoses, and cables shall be supported to prevent damage caused by vibration, chafing, or sharp bends.

3.21.1 Castings and forgings. All parts, components, and assemblies of the pump which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.21.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.21.3 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the pump to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pump shall be at the option of the contractor.

3.21.4 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS B3.0 - Welding Procedures and Performance Qualifications.

ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.21.5 Brazing or soldering. Brazing or soldering shall not be attempted as repair measures for defective parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article pump shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pump shall be subjected to tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S4.

4.4.2 Examination. Each pump shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Test.

4.4.3.1 Individual. Each pump shall be subjected to tests marked "X" in column 2 of table I. Failure of a test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to tests marked "X" in column 3 of table I. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pump shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Components missing or not as specified.
103. Nonsparking metals not used in hazardous locations.
104. Safety devices or guards missing or not as specified.
105. Major assemblies and installed attachment not as specified.
106. Nonadjustable means not provided for cooling the pumpseal with the liquid delivered by the pump.
107. Lubrication fittings not accessible or not as specified.
108. Drain outlets not accessible.
109. Identification marking missing, incomplete or illegible.
110. Workmanship not as specified.

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4.5.2 Tests.

4.5.2.1 Test conditions. Prior to tests, the pump shall be lubricated with oils and grease specified in 3.14.1 and 3.14.5. Oils shall be those designated for use in the ambient temperature at which the tests are being performed.

4.5.2.2 Test schedule.

TABLE I. Test schedule.

First article	Quality conformance: Individual	Quality conformance: Sample	Tests	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	-	-	Environmental.	4.5.2.3	3.5
X	-	-	Safety.	4.5.2.4	3.6
X	-	-	Human factors.	4.5.2.5	3.7
X	-	-	Noise level Measurement.	4.5.2.6	3.7.1
X	-	-	Reliability.	4.5.2.7.	3.8
X	-	-	Maintainability.	4.5.2.8	3.9
X	X	-	Pump and pump-shaft seal hydrostatic pressure.	4.5.2.9	3.10.1
X	X	-	Nonreverse rachet-type mechanism.	4.5.2.10	3.11.1
X	-	X	Performance.	4.5.2.11	3.12
X	-	X	Head capacity.	4.5.2.11.1	1.2
X	-	X	Horsepower requirements.	4.5.2.11.2	3.11.2
X	-	X	Surface discharge, open bypass.	4.5.2.11.3	3.10.2.4
X	-	-	Electromagnetic interference.	4.5.2.12	3.13
X	-	-	Lifting provisions.	4.5.2.13	3.16
X	-	-	Tiedown provisions.	4.5.2.14	3.17

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4.5.2.3 Environmental.

4.5.2.3.1 High temperature. The pump shall be tested as specified in MIL-STD-810, method 501.1, procedure I. The maximum ambient storage temperature shall be plus 160° F, and the maximum ambient operating temperature shall be plus 125° F. The operating period of procedure I shall be 2 hours. The pump shall be operated at rated capacity. Non-conformance to 3.5 shall constitute failure of this test.

4.5.2.3.2 Low temperature. The pump shall be tested as specified in MIL-STD-810, method 502.1, procedure I. The minimum ambient storage temperature shall be minus 50° F. The minimum ambient operating temperature shall be minus 25° F. The operating period of procedure I shall be 2 hours. The pump shall be operated at rated capacity. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.4 Safety. The pump shall be evaluated throughout all tests. Non-conformance to 3.6 shall constitute failure of this test.

4.5.2.5 Human factors. The human factors requirements shall be evaluated during testing. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.6 Noise level measurement.

4.5.2.6.1 Personnel exposure. Measure the noise level at the control panel, other operator positions, as well as any normally or occasionally occcupied positions, as specified in 5.1.2 of MIL-STD-1474. These measurements shall be made at two-thirds and at full rated load as specified in 1.2. The test surface shall be level with no sound reflecting surfaces within 100 feet. Failure to meet 3.7.1 constitutes failure of this test.

4.5.2.6.2 Exterior. Measure the noise level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.7.1 shall constitute failure of this test.

4.5.2.6.3 Hazard precautions. Measure the noise level at all control and maintenance positions while operating at the maximum noise condition for those locations. Should the noise level exceed 85 dB(A) at any position, failure to provide the precautions specified in 3.7.1 shall constitute failure of this test.

4.5.2.7 Reliability. The pump shall be tested as specified in 4.5.2.11 with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan IIC using the MTBF specified in 3.8. A failure is defined as any malfunction which cannot be corrected within 30 minutes by adjustment, repair, or replacement action using controls, maintenance tools or repair parts furnished with the pump(s) and which causes or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pump by continued operation.
c. Safety hazard(s) to personnel.

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4.5.2.8 Maintainability.

4.5.2.8.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.9. Failure of the support elements to maintain the pump as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in accessing the maintenance ratio or the MTBF.

4.5.2.8.2 Maintenance ratio. The maintenance ratio shall be computed during initial production testing. Nonconformance to 3.9.2 shall constitute failure of this test.

4.5.2.9 Pump and pump-shaft seal hydrostatic pressure. With pump-shaft installed, subject pump and pump-shaft seal to an internal hydrostatic pressure of not less than 350 psi. Leakage, failure, or permanent deformation of the pump or pump-shaft seal shall constitute failure of this test.

4.5.2.10 Nonreverse, ratchet-type mechanism. Connect the pump to the electrical-power system specified in 3.11.4 to determine whether the motor-shaft and pump-shaft rotate in the direction specified in 3.11. Stop the unit and reverse motor rotation. Operate the pump to determine whether the ratchet-type mechanism is disengaging the pump-shaft from the motor-shaft. Inability of the ratchet-type mechanism to disengage the pump-shaft from the motor-shaft when the motor-shaft rotates in the direction opposite to that specified in 3.11.1 shall constitute failure of this test.

4.5.2.11 Performance. The pump shall be tested while pumping hydrocarbons or water. If water is used as a test liquid, the maximum brake horsepower required for pumping water shall be corrected to the maximum brake horsepower required for pumping liquid hydrocarbon fuels. The specific gravity of hydrocarbons shall be between 0.68 to 0.935 at a temperature between 40° F and 80° F. The tests shall be conducted with a sustained pumping liquid level located between the centerline of the second-stage impeller and the top of the bowl of the second-stage impeller. When specified (see 6.2), a data sheet indicating the efficiency, head-capacity, and brake horsepower required by the pump; and the temperature and specific gravity of the pumped liquid shall be provided. Data points for the head-capacity curve shall be taken at increments of 20 gpm. Test data shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury and a temperature of 68° F. After the tests have been conducted, the pump shall be drained and thoroughly dried.

4.5.2.11.1 Head capacity. The head-capacity curve data shall be obtained with the pump operating at 1800 rpm. Data relative to head-capacity shall be taken at 20 gpm increments from wide open discharge to shutoff and shall include the rating point (390 feet total discharge head). Inability of the pump to attain the applicable capacity specified in 1.2 at a 390-foot discharge head shall constitute failure of this test.

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4.5.2.11.2 Horsepower requirements. Operate the pump as specified in 4.5.2.11.1 while measuring pump required horsepower to determine conformance to 3.11.2 Compute pump efficiency at all data points recorded to determine conformance to 3.12. Inability of the pump to satisfy the requirements of 3.11.2 and 3.12 shall constitute failure of the test.

4.5.2.11.3 Surface-discharge, open bypass. Operate the pump as specified in 4.5.2.11.1. Increase the total head of the pump until shutoff head is reached. After the shutoff head has been reached, operate the pump to determine conformance to 3.10.2.4. Inability to discharge a minimum of 2 percent of the rated capacity shall constitute failure of this test.

4.5.2.12 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.13. The contractor shall furnish the contracting officer the report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.13 Lifting provisions. Subject each lifting attachment to the load specified in 3.16. Failure or permanent deformation of any attachment or interference of a sling with any part of the pump shall constitute failure of this test.

4.5.2.14 Tiedown provisions. Subject each tiedown attachment to the thrusts specified in 3.17. Failure or permanent deformation of any attachment shall constitute failure of this test.

4.6 Inspection comparison. The Government may select pumps at any time during the contract production period and subject these pumps to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pumps selected at random from those which have been accepted by the Government and will not include the previously inspected first article pumps. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of any inspection comparison pump to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison pump(s) and shall be cause for refusal by the Government to continue acceptance of production pumps until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumps previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pumps accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Inspection of packaging. The preservation, packging, and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-P-710. Commercial packing shall be inspected for conformance to MIL-STD-1188.

5. PACKAGING

5.1 Preservation, packing, and marking. The degree of preservation shall be level A, level B, or commercial, as specified (see 6.2). The degree of packing shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Military levels of protection. Preservation, packing, and marking for level A or level B shall be in accordance with MIL-P-710.

5.1.2 Commercial protection. Preservation, packing, and marking for commercial protection shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The pumps are intended for use in pumping liquid hydrocarbon fuels from underground tanks.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDiss applicable and exceptions thereto (see 2.1.1).
c. Capacity of pump required (see 1.2).
d. Time frame required for submission of a first article and number of pumps required (see 3.2).
e. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. Distance from bottom of pump mounting flange to the bottom of the suction strainer (see 3.10).
g. Material for mounting flange if other than as specified and whether an integral or separate flange shall be furnished (see 3.10.2.1).
h. Impeller material required if other than as specified (see 3.10.8).
i. Impeller shaft material required if other than as specified (see 3.10.9).
j. Voltage and frequency required if other than as specified (see 3.11.4).
k. When electromagnetic interference characteristics shall conform to MIL-STD-461 (see 3.13).
l. When welding process is not at the contractor's option (see 3.21.3).
m. When data sheet is required (see 4.5.2.11).
n. Degree of preservation and degree of packing required (see 5.1).

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6.3 First article. Any changes or deviations of the production pump from the approved first article during production will be subject to approval by the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumps conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of each pump.

6.6 Noise limits. When the noise level of the pump exceeds 85dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the pump unit, the distance from the pump unit at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

6.7 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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Preparing activity:
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Project 4320-0283

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-13386D	Pumps, Centrifugal, Multistage, Vertical, Electric-Motor-Driven (for Liquid Hydrocarbon Fuels)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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